1.4.3 ▶ ▶ ▶ Parallel Axis Theorem

The Parallel Axis Theorem provides a way to find the moment of inertia of a rigid body about an axis parallel to an axis that passes through the center of mass. First, consider the following rigid body with moment of inertia I_{cm} about the center of mass. Further consider that we wish to find the moment of inertia about the axis perpendicular to the page and passing through point O.

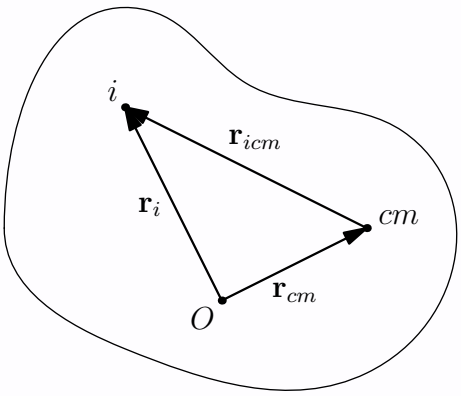

Fig. 1.11

By the definition of moment of inertia, the moment of inertia of the body around O is $I_O = \sum m_i r_i^2$. Writing r_i as a sum of vectors, we have $r_i = r_{icm} + r_{cm}$. Plugging into our definition yields

$$I_O = \sum m_i r_{icm}^2 + \sum m_i r_{cm}^2 + 2r_{cm} \cdot \sum m_i r_{icm}. \tag{1.60}$$

The first term is simply equal to I_{cm} by definition. The second term is equal to Mr_{cm}^2 since r_{cm} is a constant. The third term equals zero due to the definition of center of mass[7]. Thus, we have the parallel axis theorem, which states that the moment of inertia of a rigid body along an axis parallel to an axis through the center of mass equals the moment of inertia of the center of mass times Mr_{cm}^2 where r_{cm} is the distance between the axes. That is,

$$I_O = I_{cm} + Mr_{cm}^2 \tag{1.61}$$

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[7]See sec. 1.3.5 for a more thorough explanation